FINITENESS OF COHOMOLOGY OF A SHEAF OF MODULES ON A CURVE IS PRESERVED BY FINITE PUSHFORWARDS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the vanishing of the higher direct images, the Leray identification, and the transfer of finiteness.

Editorial note. The label lemma-finite-pushforward-coefficients names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

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Allowed standard prerequisites (editorial).

- Vanishing of the higher direct images of a finite morphism. Same chapter, the proposition stating that a finite morphism has vanishing higher direct images. Supplies R^q f_* = 0 for q > 0.
- The Leray spectral sequence in the form used for a finite morphism. Cohomology on Sites, the lemma applying Leray, which for a finite morphism degenerates. Supplies the identification of the cohomology of X with that of Y and f_*F.

Source reference. Original reference 03QP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8245–8245. Prerequisite pointer invoked at proof line(s) 15921; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0733: sources/stacks-chow/tags.txt, source lines 9047–9047. Prerequisite pointer invoked at proof line(s) 15923; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let Λ be a Noetherian ring, let k be an algebraically closed field, let $f : X \to Y$ be a finite morphism of separated finite type schemes over k of dimension ≤ 1, and let $\mathcal{F}$ be a sheaf of Λ-modules on $X_{étale}$. If $H^q_{étale}(X, \mathcal{F})$ is a finite Λ-module, then so is $H^q_{étale}(Y, f_*\mathcal{F})$.*

Proof. Namely, we have $H^q_{étale}(X, \mathcal{F}) = H^q_{étale}(Y, f_*\mathcal{F})$ by the vanishing of $R^q f_*$ for $q > 0$ (Proposition 03QP) and the Leray spectral sequence (Cohomology on Sites, Lemma 0733). □

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